

Homework problems:

Exercise 9.1 (15 points). Solve all the exercises shown on the page Problem H.9.1 - Real Estate Prices until June 15, 09:00.

Remarks:

1. The points for this problem count for the grade bonus. Hence, you will have to upload your solution to Moodle until **June 15, 09:00**. Use the link given on Moodle under the topic "Peer-Review of Homework Problems" to upload your solution.

Your upload consists of the Quarto file containing your solution **and** the HTML file, that you generated by rendering the Quarto file.

Note that you **SHOULDN'T** write your name on your submission. You are identified by signing in on Moodle.

2. You can earn up to **five additional points**. To do this, you need to grade three solutions (your own and two from your peers) that will be assigned to you on Moodle for evaluation. You can gain up to five additional points based on the grading quality. See next week's problem set for more details.

3. Note that there is a template file `starter-real_estate.qmd` on Moodle for this problem (in the folder *Data, Quarto and R files - Sample solutions and previous problem sets*). You can certainly change the name of the file, but don't choose your name as file name.

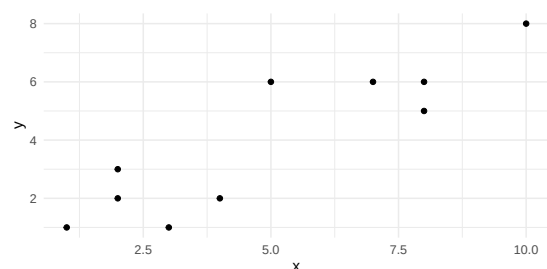
In case you don't use the template file, make sure to include in the YAML header the option `embed-resources: true`:

```
format:
  html:
    embed-resources: true
```

Solution: Check again the page Problem H.9.1 - Real Estate Prices to find the solution to the problem as well as some **grading instructions**.

Exercise 9.2. A medication prescribed for depression is under investigation for its potential to reduce response capacity as a side effect. In this study, a clinic selected ten patients at random to receive varying doses of the medication.

Response capacity was measured by instructing each patient to press a button immediately upon receiving a specific signal. The interval between the signal and the button press served as the measure of response capacity. The resulting data include the medication dosage X (in mg) and the corresponding response time Y (in seconds).



- a) What does the scatter plot indicate about the relationship between X and Y ?

- b) Calculate the fitted regression line. Assess the model fit.
- c) Give an interpretation of the slope estimate.
- d) A patient has been administered a dose of 5.5 mg of the medication. What reaction time is expected based on the fitted regression model?

Hint:

```
c(sum(df$x), sum(df$x^2), sum(df$y), sum(df$y^2), sum(df$x * df$y))
```

```
[1] 50 336 40 216 262
```

Solution:

- a) The scatter plot demonstrates a strong positive correlation, indicating a linear relationship between the variables. The data suggest that reaction time increases as dosage increases.
- b) We want to estimate the intercept β_0 and the slope β_1 of the regression line $y = \beta_0 + \beta_1 \cdot x$.
From the lecture notes, we know that the least-squares estimators of β_1 and β_0 are given by

$$\hat{\beta}_1 = \frac{s_y}{s_x} r_{(x,y)} \quad \text{and} \quad \hat{\beta}_0 = \bar{y}_n - \hat{\beta}_1 \bar{x}_n.$$

Also, from the lecture notes, we know that

$$r_{(x,y)} = \frac{\frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x}_n)(y_i - \bar{y}_n)}{s_x \cdot s_y} =: \frac{s_{xy}}{s_x \cdot s_y},$$

where s_{xy} is called the empirical covariance between x and y .

First, we try to rewrite the estimator $\hat{\beta}_1$, such that we can take the values from the hint:

$$\hat{\beta}_1 = \frac{s_y}{s_x} r_{(x,y)} = \frac{s_y}{s_x} \cdot \frac{s_{xy}}{s_x \cdot s_y} = \frac{s_{xy}}{s_x^2}.$$

But now we realize that that hint does not contain $s_{xy} = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x}_n)(y_i - \bar{y}_n)$. It only contains $\sum_{i=1}^n x_i \cdot y_i$. It also does not contain the values for s_x^2 and s_y^2 . But for them, we can use the formula

$$s_x^2 = \frac{1}{n-1} \cdot \left(\sum_{i=1}^n x_i^2 - n \bar{x}_n^2 \right),$$

derived in Chapter 5.3. Therefore, let's try to rewrite s_{xy} in the same way.

$$\begin{aligned} s_{xy} &= \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x}_n)(y_i - \bar{y}_n) \\ &= \frac{1}{n-1} \left(\sum_{i=1}^n x_i \cdot y_i - \sum_{i=1}^n \bar{x}_n \cdot y_i - \sum_{i=1}^n x_i \cdot \bar{y}_n + \sum_{i=1}^n \bar{x}_n \cdot \bar{y}_n \right) \\ &= \frac{1}{n-1} \left(\sum_{i=1}^n x_i \cdot y_i - \bar{x}_n \cdot n \cdot \bar{y}_n - n \cdot \bar{x}_n \cdot \bar{y}_n + n \cdot \bar{x}_n \cdot \bar{y}_n \right) \\ &= \frac{1}{n-1} \left(\sum_{i=1}^n x_i \cdot y_i - n \cdot \bar{x}_n \cdot \bar{y}_n \right) = \frac{1}{n-1} \sum_{i=1}^n x_i \cdot y_i - \frac{n}{n-1} \cdot \bar{x}_n \cdot \bar{y}_n. \end{aligned}$$

Using this result and the values from the hint, we obtain

$$\hat{\beta}_1 = \frac{s_{xy}}{s_x^2} = \frac{\frac{1}{n-1} \sum_{i=1}^n x_i y_i - \frac{n}{n-1} \bar{x}_n \bar{y}_n}{\frac{1}{n-1} \sum_{i=1}^n x_i^2 - \frac{n}{n-1} \bar{x}_n^2} = \frac{\frac{1}{9} \cdot (262 - 10 \cdot \frac{1}{10} \cdot 50 \cdot \frac{1}{10} \cdot 40)}{\frac{1}{9} \cdot (336 - 10 \cdot (\frac{1}{10} \cdot 50)^2)} \approx 0.72$$

and

$$\hat{\beta}_0 = \bar{y}_n - \hat{\beta}_1 \bar{x}_n = \frac{1}{10} \cdot 40 - \hat{\beta}_1 \cdot \frac{1}{10} \cdot 50 \approx 0.40.$$

For the R-squared value R^2 , we get

$$\begin{aligned} R^2 &= r_{(x,y)}^2 = \left(\frac{s_{xy}}{s_x s_y} \right)^2 = \left(\frac{s_{xy}}{s_x^2} \cdot \frac{s_x}{s_y} \right)^2 \\ &= \left(\hat{\beta}_1 \cdot \frac{s_x}{s_y} \right)^2 \approx \left(0.72 \cdot \frac{\sqrt{\frac{1}{9} \cdot (336 - 10 \cdot (\frac{1}{10} \cdot 50)^2)}}{\sqrt{\frac{1}{9} \cdot (216 - 10 \cdot (\frac{1}{10} \cdot 40)^2)}} \right)^2 \\ &= 0.7961143. \end{aligned}$$

Thus, approximately 80% of the total variance is accounted for by the regression model, indicating a strong fit.

- c) For a one-unit increase in dosage, the reaction time will, on average, increase by 0.72 seconds.
- d) For a patient with a dosage of 5.5 mg, the reaction time of the patient according to the model is

$$0.40 + 0.72 \cdot 5.5 = 4.36$$

seconds.

Exercise 9.3. Suppose we have a data set with five predictors, $X_1 = \text{GPA}$, $X_2 = \text{IQ}$, $X_3 = \text{Level}$ (1 for College and 0 for High School), $X_4 = \text{Interaction between GPA and IQ}$, and $X_5 = \text{Interaction between GPA and Level}$. The response is the starting salary after graduation (in thousands of dollars). Suppose we use least squares to fit the model, and get $\hat{\beta}_0 = 50$, $\hat{\beta}_1 = 20$, $\hat{\beta}_2 = 0.07$, $\hat{\beta}_3 = 35$, $\hat{\beta}_4 = 0.01$, and $\hat{\beta}_5 = -10$.

- a) Identify the correct answer and provide a justification.
 - i) For a fixed value of IQ and GPA, high school graduates earn more on average than college graduates.
 - ii) For a fixed value of IQ and GPA, college graduates earn more on average than high school graduates.
 - iii) For a fixed value of IQ and GPA, high school graduates earn more on average than college graduates, provided that the GPA is high enough.
 - iv) For a fixed value of IQ and GPA, college graduates earn more on average than high school graduates, provided that the GPA is high enough.
- b) Predict the salary of a college graduate with an IQ of 110 and a GPA of 4.0.
- c) Determine whether the following statement is true, false, or undecidable, and provide justification.

Since the estimated coefficient for the GPA/IQ interaction term is much smaller than the one obtained for the GPA/Level interaction, the GPA/IQ interaction is significantly less important for predicting the starting salary than the GPA/Level interaction.

Solution:

- a) The fitted model has the form:

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 \cdot \text{GPA} + \hat{\beta}_2 \cdot \text{IQ} + \hat{\beta}_3 \cdot \text{Level} + \hat{\beta}_4 \cdot \text{GPA} \cdot \text{IQ} + \hat{\beta}_5 \cdot \text{GPA} \cdot \text{Level}.$$

To answer the question, we need to fix IQ and GPA and compare the average earnings of high school graduates with those of college graduates, i.e., changing the Level from 0 to 1. Subtracting the average salary of high school graduates from those of college graduates leads to

$$\begin{aligned} \Delta y &= \hat{\beta}_0 + \hat{\beta}_1 \cdot \text{GPA} + \hat{\beta}_2 \cdot \text{IQ} + \hat{\beta}_3 \cdot \text{Level} + \hat{\beta}_4 \cdot \text{GPA} \cdot \text{IQ} + \hat{\beta}_5 \cdot \text{GPA} \cdot \text{Level} \\ &\quad - (\hat{\beta}_0 + \hat{\beta}_1 \cdot \text{GPA} + \hat{\beta}_2 \cdot \text{IQ} + \hat{\beta}_4 \cdot \text{GPA} \cdot \text{IQ}) \\ &= \hat{\beta}_3 + \hat{\beta}_5 \cdot \text{GPA}. \end{aligned}$$

- i) For this answer to be correct, we need $\Delta y < 0$. But this is only true if

$$0 > \Delta y = 35 - 10 \cdot \text{GPA} \Leftrightarrow \text{GPA} > 3.5,$$

and not for any fixed GPA value. So, the answer is wrong.

- ii) For this answer to be correct, we need $\Delta y > 0$. But this is only true if

$$0 < \Delta y = 35 - 10 \cdot \text{GPA} \Leftrightarrow \text{GPA} < 3.5,$$

and not for any fixed GPA value. So, the answer is wrong.

- iii) From i) we know that high school graduates with a GPA over 3.5 will have a higher starting salary on average. So, the answer is correct.

- iv) Since iii) is correct, this answer must be wrong.

- b) Evaluating the fitted model with the given predictor values yields the following results:

$$\begin{aligned} \hat{y} &= \hat{\beta}_0 + \hat{\beta}_1 \cdot 4.0 + \hat{\beta}_2 \cdot 110 + \hat{\beta}_3 \cdot 1 + \hat{\beta}_4 \cdot 4.0 \cdot 110 + \hat{\beta}_5 \cdot 4.0 \\ &= 50 + 20 \cdot 4.0 + 0.07 \cdot 110 + 35 + 0.01 \cdot 440 - 10 \cdot 4.0 \\ &= 137.1. \end{aligned}$$

- c) The statement is false — and here's why: The magnitude of a regression coefficient alone does not determine the importance of a predictor or interaction term. The coefficient size is influenced by the scale of the variables involved. For example, IQ might range from 80 to 160, while Level is binary (0 or 1). So even a small coefficient on $\text{GPA} \cdot \text{IQ}$ could have a large effect if the values of GPA and IQ are large. As an example, assume GPA is 4 and IQ is 120. The interaction term then contributes:

$$0.01 \cdot 4 \cdot 120 = 4.8,$$

which is a non-negligible contribution to the predicted salary.

Remark: To determine which interaction is more important, we would need to apply a statistical test or at least do a model comparison using a metric like the adjusted R^2 , not just the raw coefficients. As a result of such a comparison, it might be concluded that the interaction term for GPA/IQ is less important than the one for GPA/Level. However, the given statement is still incorrect because it suggests that the importance can be determined solely based on the size of the estimated coefficient.